Tidal potential measured on a coastal–inland optical-clock baseline

Introduction

Measurements of Earth's gravity field reveal how water, air and the solid Earth redistribute mass. Ground gravimetry measures gravitational acceleration and its spatial gradients, while precise levelling combined with gravity observations determines potential differences.[1,2] Satellite missions have tracked large-scale changes in water storage and the climate system.[3,4] Monitoring time-varying potential differences between distant sites requires resolving weak signals and distinguishing contributions with similar temporal signatures.[5] Tides provide a well-defined setting for this challenge. The Moon's and Sun's tidal fields act together with deformation of the solid Earth and redistribution of ocean water. Atmospheric pressure tides also force the ocean.[6] Solid-Earth tidal potentials can reach amplitudes of several $\mathrm{m}^2/\mathrm{s}^2$ at individual sites, while resolving smaller contributions requires greater measurement sensitivity.[7]

Optical clocks offer a further way to measure potential differences through their relative ticking rates. Near Earth's surface, a fractional frequency change of 10^{-18} corresponds to about 1 cm of equivalent height.[8] Experiments have resolved gravitational redshifts on laboratory scales and measured static potential differences between distant sites.[2,9,10,11] Optical clocks now reach fractional systematic uncertainties near 10^{-19}, and separate lattice-clock experiments have achieved instabilities of a few $10^{-17}/\sqrt{\tau/\mathrm{s}}$, where τ is averaging time.[12,13] Non-tidal mass changes in the atmosphere, oceans and continental water can matter at the 10^{-18} fractional-frequency level.[7] Optical-fibre networks and advances in free-space frequency transfer are extending clock comparisons over long distances.[14,15,16,17,18] These advances motivate optical-clock networks for monitoring time-varying gravitational potential.[5,8] Resolving neighbouring tidal components, however, requires sufficient sensitivity and temporal coverage, particularly when observations are intermittent.

Here we detect tidal variations in the potential difference between a strontium clock in coastal Shanghai and a ytterbium clock in inland Wuhan. The clocks are separated by 692 km and connected by a 1,350-km fibre link.[19] The combined solid-Earth and ocean-loading prediction gives an M_2 fractional-frequency amplitude of 4.78×10^{-18}, equivalent to about 4.4 cm of height.[20,21] We use 545.9 hours of observations spread over 353 days. Using the predictable tidal phases, we combine variations measured within individual observing segments while allowing an independent frequency offset for each segment. The changing relative phase of the lunar and solar tides enables joint estimation of the responses of two neighbouring semidiurnal tidal groups, M_2+N_2 and S_2+K_2, with modelled phases and relative amplitudes fixed within each group.

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